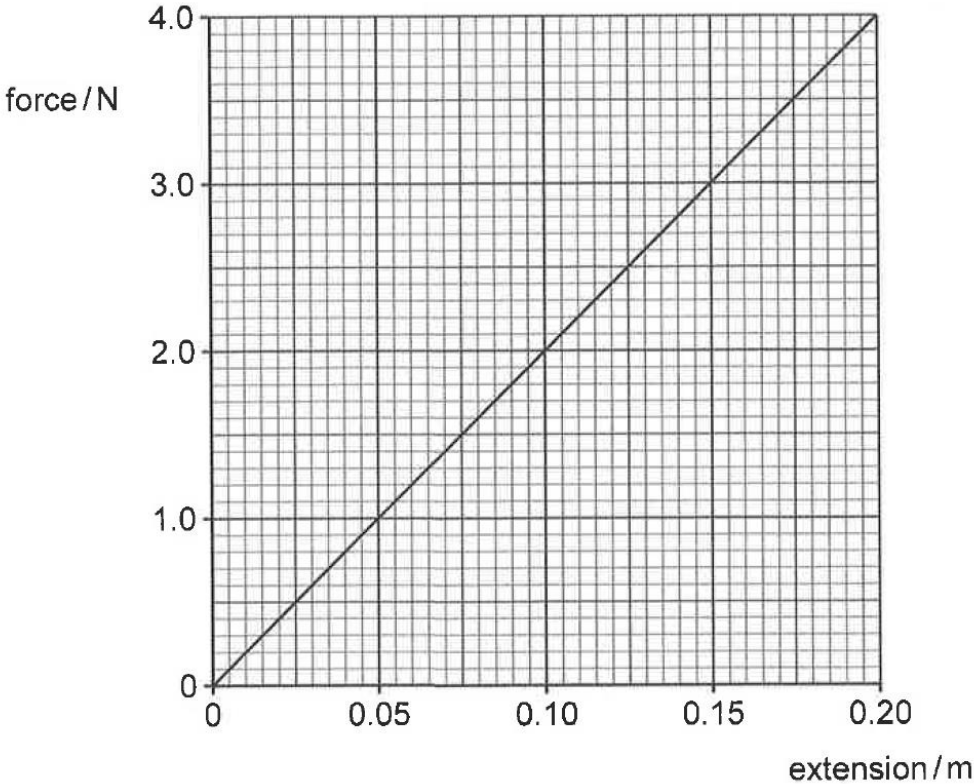
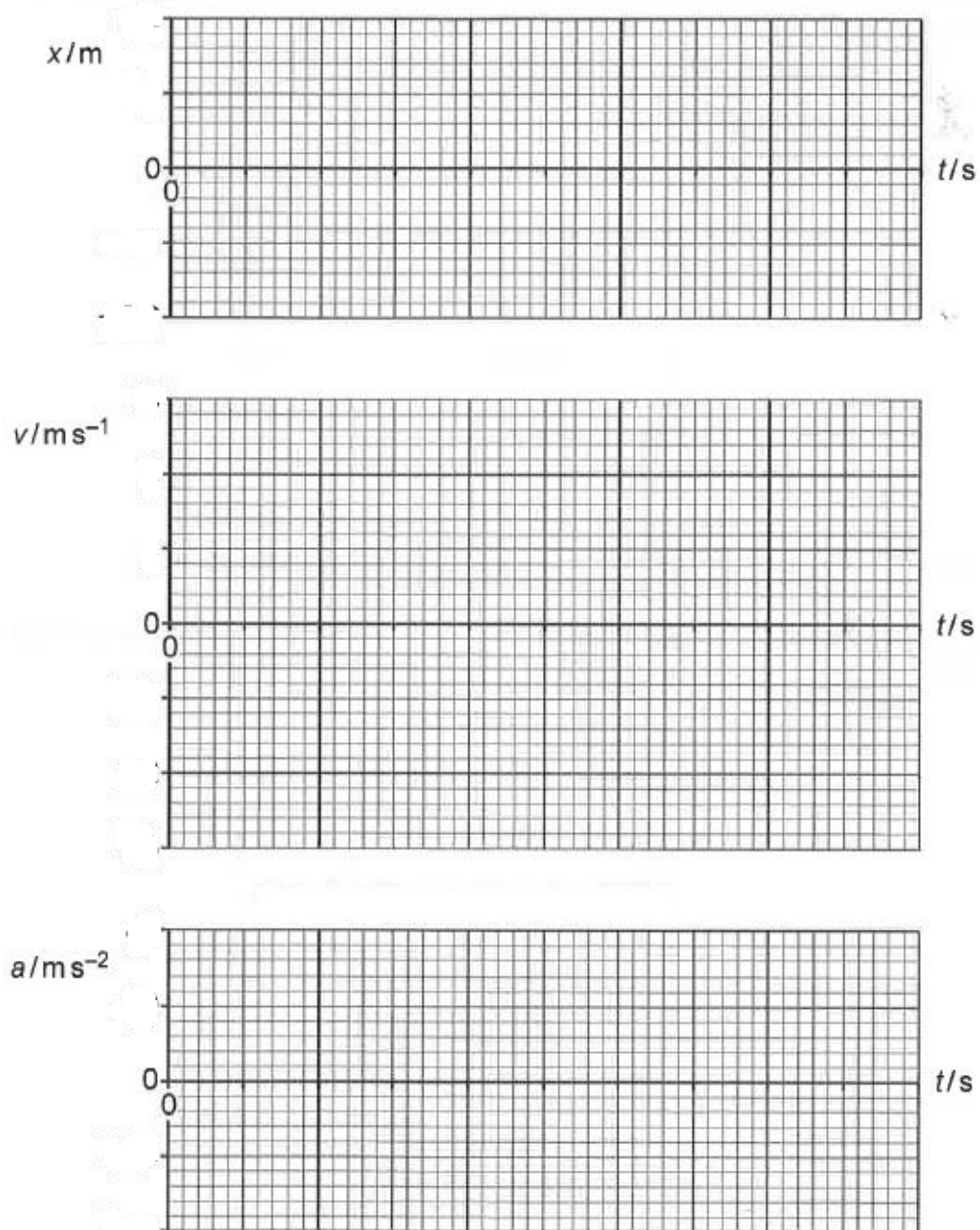


8	(a)	Define <i>simple harmonic motion</i>	
			
			
			[2]
		Simple harmonic motion is a periodic motion in which the <u>acceleration is proportional to the displacement from the equilibrium position</u> And <u>directed towards the equilibrium position</u> OR <u>in the opposite direction to the displacement.</u>	B1 B1
	(b)	Fig. 8.1 shows the force–extension graph for a light spring.  Fig 8.1 The spring described by Fig. 8.1 is attached to a fixed point on the ceiling and a mass of 2.0 N is hung on the spring. Once the mass reaches its equilibrium position, it is displaced a further 0.15 m downward and released, such that it oscillates with simple harmonic motion.	

		(i)	Determine the force constant k of the spring.	
			$k = \dots\dots\dots \text{N m}^{-1}$	[2]
			Solution: $k = \frac{F}{x} = \frac{4.0\text{N}}{0.20\text{m}}$ $k = 20.0 \text{ N m}^{-1}$	M1 A1
		(ii)	Show that the maximum acceleration of the mass when it is oscillating in simple harmonic motion is 14.7 m s^{-2} .	
				[3]
			Solution: <i>At amplitude position</i> $F_{\text{net}} = kx - mg \text{ [mark for setting up } F_{\text{net}} \text{ correctly]}$ $F_{\text{net}} = (20.0)(0.10 + 0.15) - 2.0$ $F_{\text{net}} = 3.0 \text{ N [mark for the correct value]}$ $ma = 3.0 \text{ N}$ $\frac{2.0}{9.81}a = 3.0 \text{ [mark for correct substitution in } ma]$ $a = 14.7 \text{ m s}^{-2}$	M1 C1 M1 A0
		(iii)	Hence, determine the period of the oscillation.	

			period = s [3]
		Solution: using $a = -\omega^2 x$ $a_0 = -\omega^2 x_0$ $-14.7 = -\omega^2 (0.15)$ $\omega^2 = \frac{14.7}{0.15} = 98$ $\omega = 9.90 \text{ rad s}^{-1} = \frac{2\pi}{T}$ $T = 0.63 \text{ s}$	M1 M1 A1
	(c)	On Fig. 8.2, sketch the variations with time of the displacement x, the velocity v and the acceleration a of the object for two complete oscillations, starting at $t = 0$. Take upwards as positive. Include an appropriate scale on the axes.	

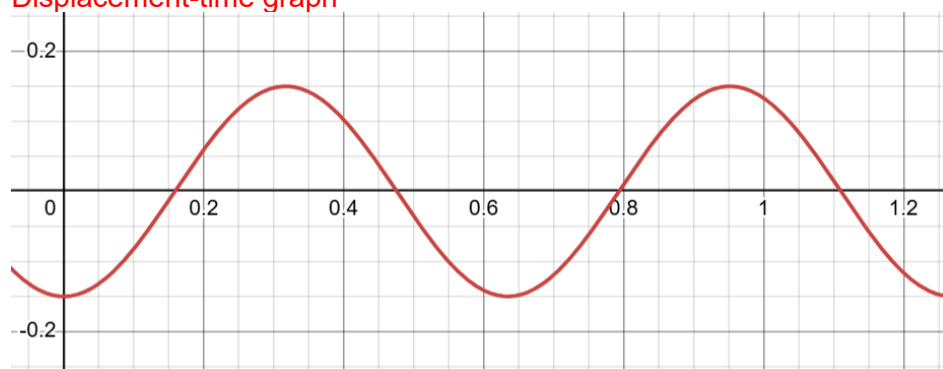


[6]

Fig. 8.2

Solution:

Displacement-time graph



[Turn over

1 mark: negative cos curve for 2 cycles (regardless of period value)
 1 mark: amplitude = 0.15 m (ecf), period = 0.63 s (ecf)

B1
 B1

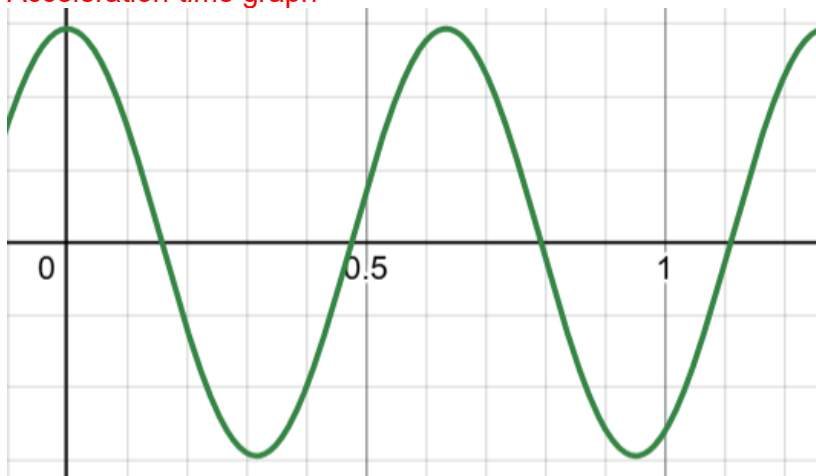
Velocity – time graph



1 mark: positive sin curve for 2 cycles (regardless of period value)
 1 mark: max $v = 1.5 \text{ m s}^{-1}$ (from $v_0 = \omega x_0 = \frac{2\pi}{0.63} \times 0.15$), period = 0.63 s
 (ecf if wrong value but same as x-t graph)

B1
 B1

Acceleration-time graph



1 mark: positive cos curve for 2 cycles (regardless of period value)
 1 mark: max $a = 14.7 \text{ m s}^{-2}$, period = 0.63 s (ecf if wrong value but same as x-t graph)

B1
 B1

(d) A second, identical spring is attached in parallel to the first spring as shown in Fig. 8.3.

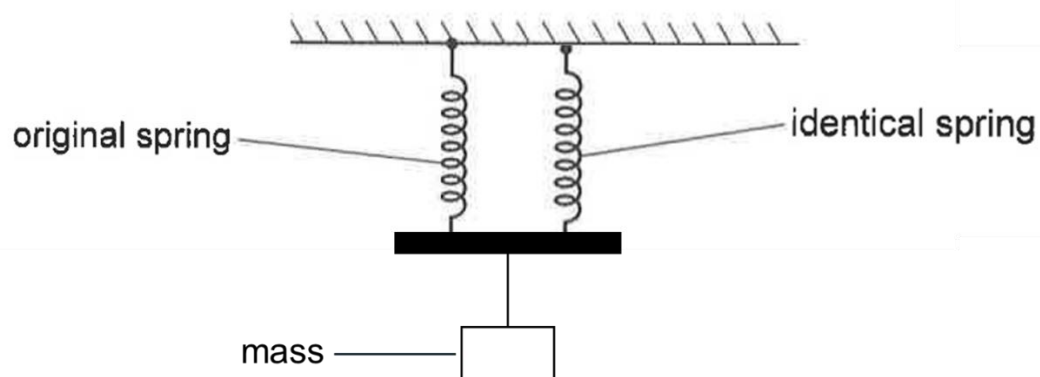


Fig. 8.3

		(i)	State and explain how the extension of the spring system compares with that of the original single spring when the same 2.0 N mass is suspended from it.
			
			
			 [2]
			Solution: Effective spring constant doubles for springs in parallel M1 Extension is halved A1
		(ii)	The mass is again displaced by 0.15 m and released to oscillate. State and explain how the period of oscillation of the new system compares with the period found in (b)(iii).
			
			
			
			
			 [2]
			Solution: Higher k means resultant force is higher, acceleration is higher and angular frequency is higher. M1 Hence, period decreases. A1

[Total: 20]

